Calibrating Temperature Sensors

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Intro

Calibration of an NTC using a DS18B20, which is a pre-calibrated temperature sensor.

The DS18B20 has 3 pins. Its digital output pin uses an open-drain architecture.

Setup

1.

The circuit is a simple voltage divider with the NTC connected on one side of the GND.

A 3.3 kΩ resistor is used to keep the output within the measurable range.

2.

Both the DS18B20 and the NTC are immersed in a water bath, close together.

The bath is continuously stirred.

3.

Middle node of the voltage divider setup is connected to an Arduino ADC pin.

The data wire of the DS18B20 is connected to a digital pin.

Next steps

Data Gathering

Sample Gathering → Three setups: 3 × DS18B20 and 3 × NTC at 3 random positions in the same water bath.

Treatments → Ice cooled the bath; a hot plate heated it. Each member monitored the 3 device pairs.

Monitoring → Temperature vs time and vs raw ADC were plotted. Each run was saved as a log and a CSV.

Response Time → NTC moved from a room-temperature bath to an ice bath until 0 °C, then back.

Results

Two ways to approach the solution

Heating and cooling runs were fitted separately first. Then we compared an empirical fit with a physical model.

Polynomial fit

Cubic is best: RMSE 0.35 °C (heating) and 0.55 °C (cooling).

NTC β-model

ln r vs 1/T is a straight line: β = 3965 K, RMSE 0.60 °C with 2 parameters.

Data used

3 valid runs: 1 heating, 2 cooling

5319 samples from 0 to 98 °C

12 rows removed: 3 empty, 9 outliers (beyond 2.5σ)

Polynomial fit

Global cubic: heating and cooling data combined

2.3 → 0.35 °C

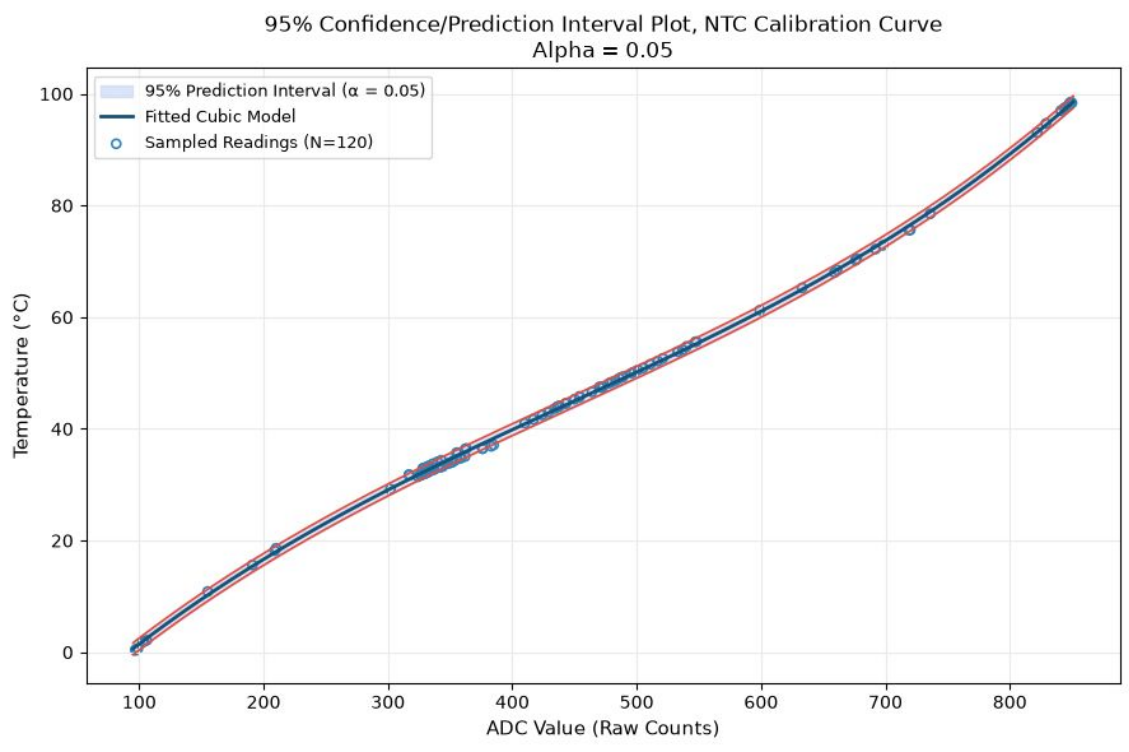
Heating RMSE, straight line vs cubic

0.54 °C

Global cubic RMSE (N = 5319)

±1.06 °C

Mean 95 % prediction half-width



$T = 1.869\times10^{-7}\,a^3 - 2.495\times10^{-4}\,a^2 + 0.2134\,a - 17.558$, a = ADC count

NTC β-model: from circuit to temperature

1

Divider

$r = (1023 - ADC) / ADC$

r = R(NTC) / R(fixed). Supply drift cancels.

2

Arrhenius

$R \propto \exp(\beta / T)$

Carriers jump the band gap: β = Eg / 2kB

3

Linear form

$\ln r = \beta \cdot (1/T) + c$

The slope of ln r vs 1/T gives β

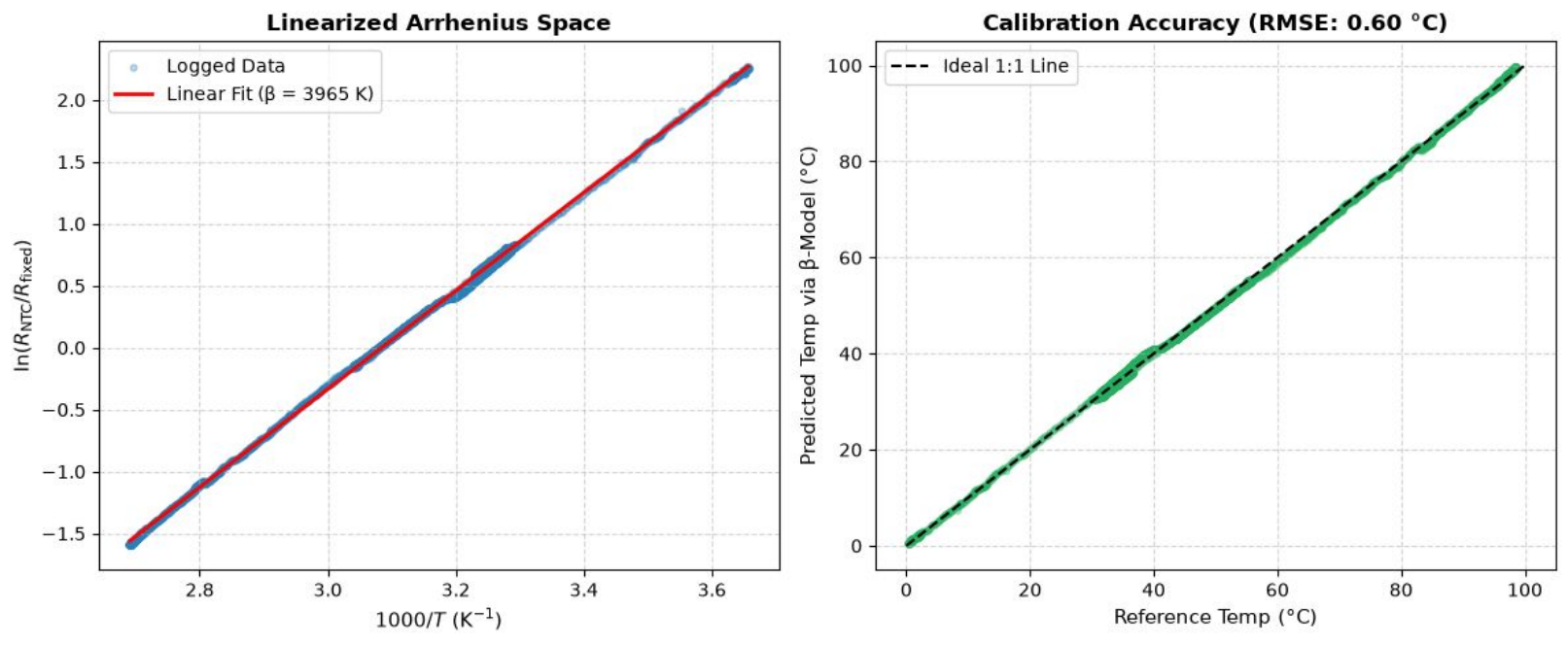
4

Runtime

$T = 1 / (A \ln r + B) - 273.15$

A = 1/β; used on the Arduino

β-model validation



A straight line of ln r vs 1/T gives β = 3965.3 K with R^2 = 0.9993 and RMSE 0.60 °C, from only two parameters.

Hysteresis?

1.16 °C

Largest heating vs cooling gap (mean 0.61 ° C)

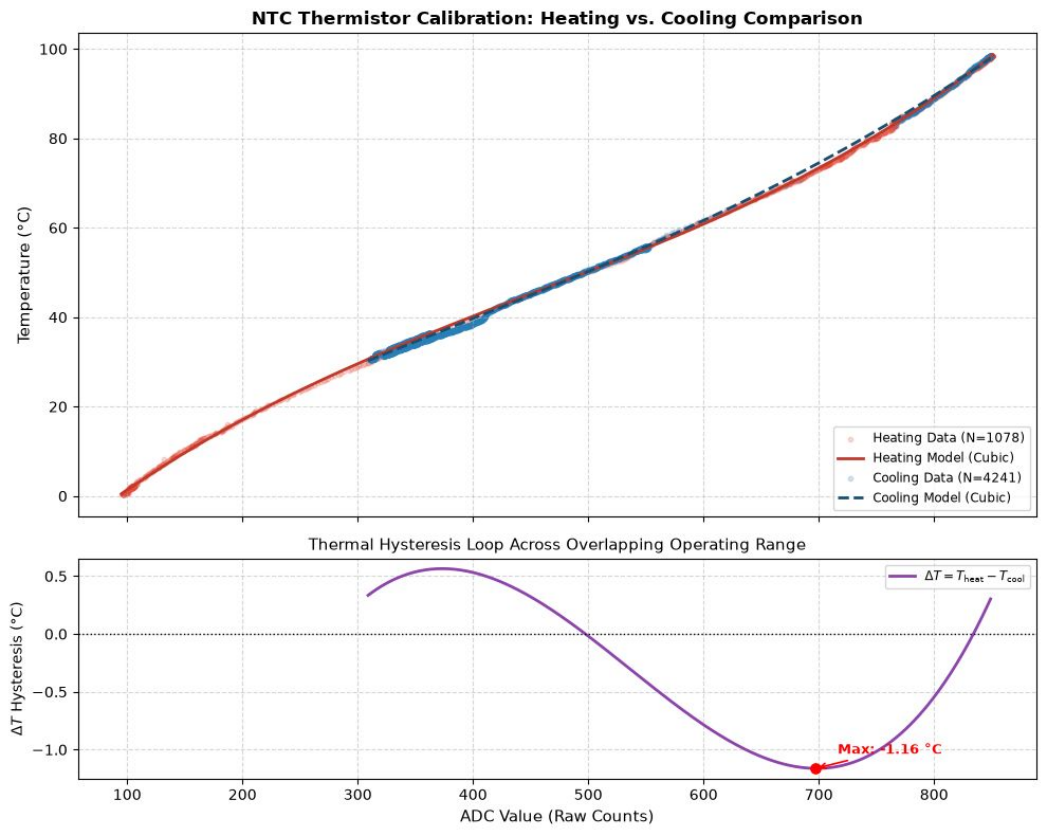
Likely causes

Thermal lag between the probes

Subtracting two cubics

Bath temperature gradients

The gap is below the reference uncertainty, so we do not claim real hysteresis.



Sensitivity

dT/dADC in °C per count. Smaller means finer resolution.

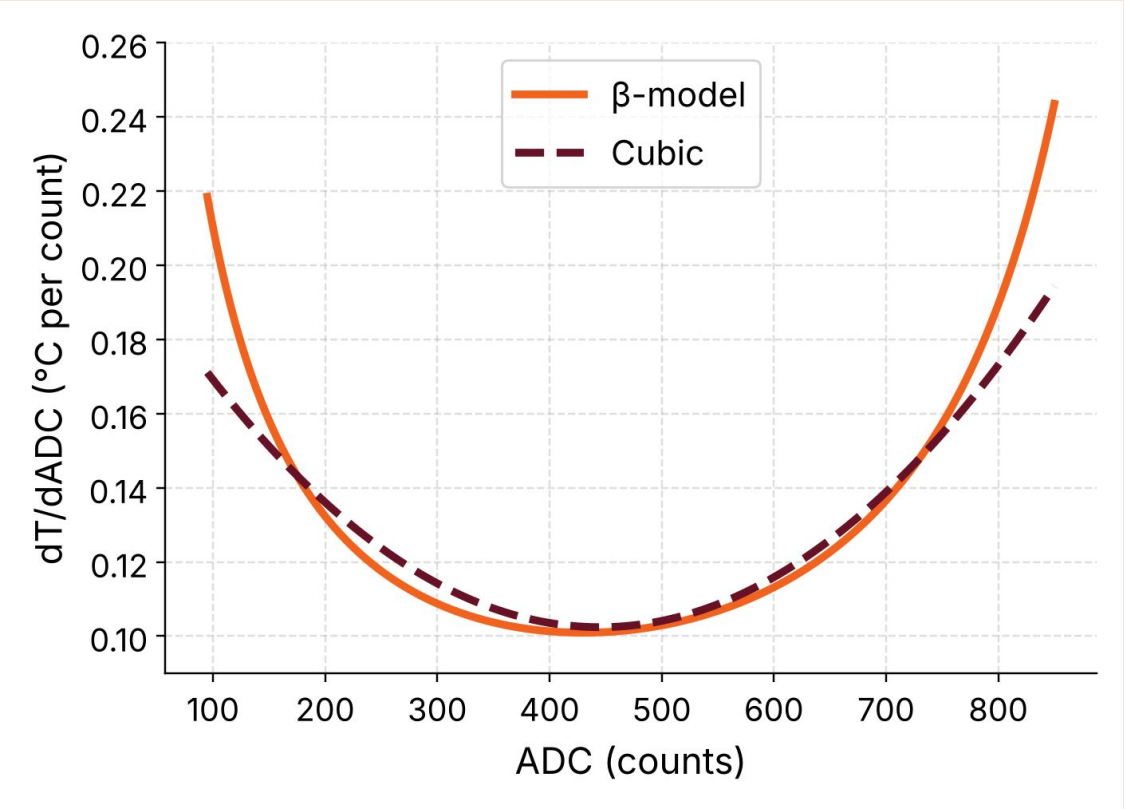
≈ 0.10 °C

Per count near mid-range (ADC ≈ 440)

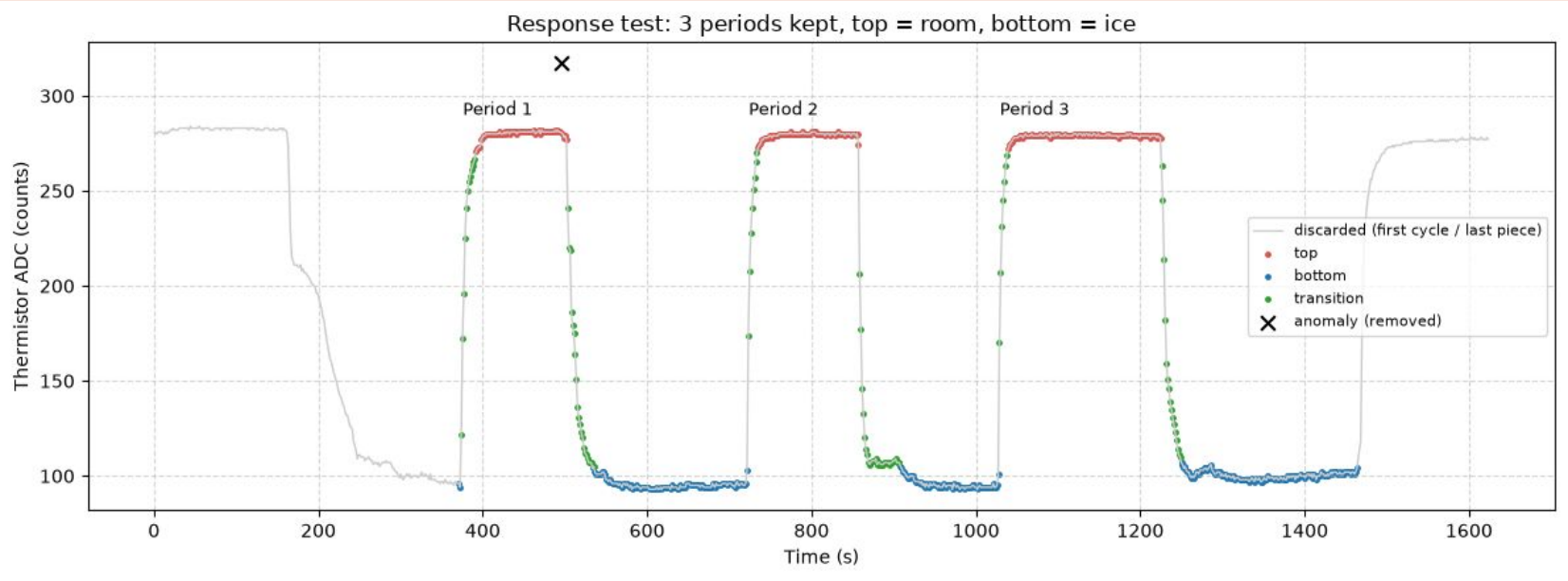
0.17–0.25 °C

Per count at the ends of the range

Both models agree: best in the middle, coarser at both ends.



Response time: the step test

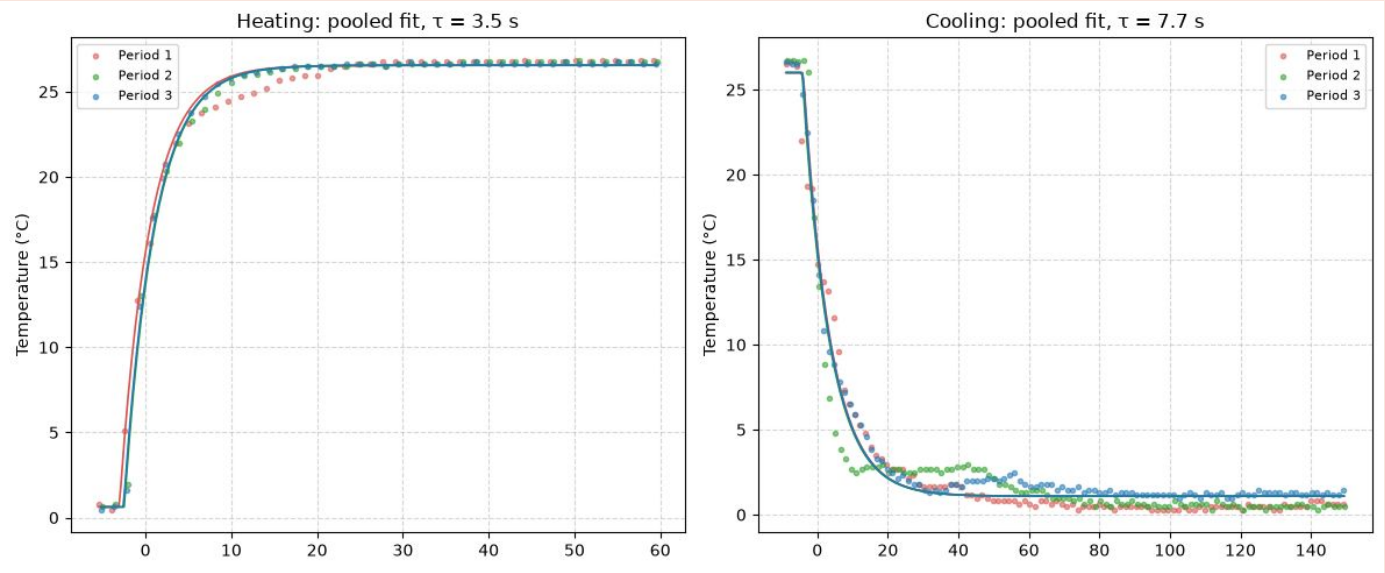


NTC moved between room-temperature water and an ice bath

3 clean periods kept; 1 spike removed

Each step fitted with a delayed first-order model

Response time: τ from the fit



Pooled fit of 3 periods, cubic calibration (β-model: τ = 3.3 s and 8.6 s).

3.5 s

τ heating, ±0.3 s

7.7 s

τ cooling, ±2.2 s

7.8 | 16.8 s

10–90 % time, heating | cooling

0.47 | 0.92 °C

Fit RMSE, heating | cooling

Uncertainty

Error Analysis

±1.06 °C

Mean 95 % prediction half-width of the global cubic

Reference → DS18B20 is ±0.5 °C. We assume about ±1 °C in practice, so smaller errors cannot be proven.

ADC steps → 10-bit ADC: each count is 0.10 to 0.25 °C, coarsest at both ends of the range.

Fit → RMSE 0.54 °C (cubic), 0.60 °C (β-model). Residuals are correlated, so standard errors are optimistic.

Process → Thermal lag and bath gradients. Self-heating was not quantified. Only 3 usable runs, 0–98 °C.

Discussion and conclusion

Two models

Cubic: 4 parameters, RMSE 0.54 °C. β-model: 2 parameters, RMSE 0.60 °C. Both are within the ±1 °C reference.

Dynamics

τ ≈ 3.5 s heating, 7.7 s cooling. The heating/cooling gap (about 0.6 °C) is below the reference uncertainty.

Recommendation

Use the β-model as the main calibration. Keep the cubic for the lowest error inside 0 to 98 °C.